

Project 2:

Network Design Rationale

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This layout centers on two logical domains with a tight control point in between. I merged Subnet 1 and Subnet 2 into Network A, so everyday office devices live on one Class C segment and share resources without extra hops. Both access switches remain in place for port capacity and physical reach, linked together and back to the Subnet_Router as the single default gateway for Network A. The Server_Router sits on the other side of a small point-to-point link and anchors Network 3, which holds the acquired company's gear and its local server. This puts a simple Layer 2 core on each side and pushes all inter-network decisions to the routers, where policy is easier to manage and audit.

Addressing is straightforward and consistent. Network A uses 192.168.1.0/24 and Network 3 uses 192.168.10.0/24, each with a 255.255.255.0 mask that leaves ample headroom while staying easy to read and troubleshoot. Router interfaces take the .1 address on each subnet, which makes the gateway obvious for users and documentation. The serial link between routers runs on a tiny 10.0.0.0/30 block, which wastes almost no space and keeps routing tables clean. Clear boundaries, no overlapping ranges, and reserved infrastructure addresses at the low end reduce guesswork and prevent the usual .0 and .255 pitfalls.

For client configuration, I centralized DHCP on Server_Main and relayed requests from Network A. The scope hands out 192.168.1.x addresses starting above the reserved static range, sets the gateway to 192.168.1.1, and supplies a working DNS resolver. Because the server lives across the routed link, the Subnet_Router uses an IP helper address to forward broadcasts to the DHCP service, which solves the "clients can't find the server" problem without extra complexity on the switches. Centralized leases make growth painless and keep conflicts off the table. Static IPs remain available for printers and other fixed devices that benefit from predictability.

Routing and translation finish the picture. I used static routes on both routers so each side knows exactly how to reach the other: Network A points 192.168.10.0/24 to the Server_Router, and Network 3 points 192.168.1.0/24 back to the Subnet_Router. Per the project requirement, I configured NAT with PAT on the Server_Router, marking the Network A interface as inside and the Network 3 interface as outside, then enabling overload on the outside interface. That choice gives every host in Network A a consistent translated source when it talks to Network 3, which simplifies any future access rules and hides internal addressing. Tests confirmed what the design set out to achieve. Devices on Network A can reach Network 3 reliably, replies come back cleanly, and the routers remain the one place where policy and visibility live.